

Q1 Answers (2 Marks)

i) Define degree distribution with an example.

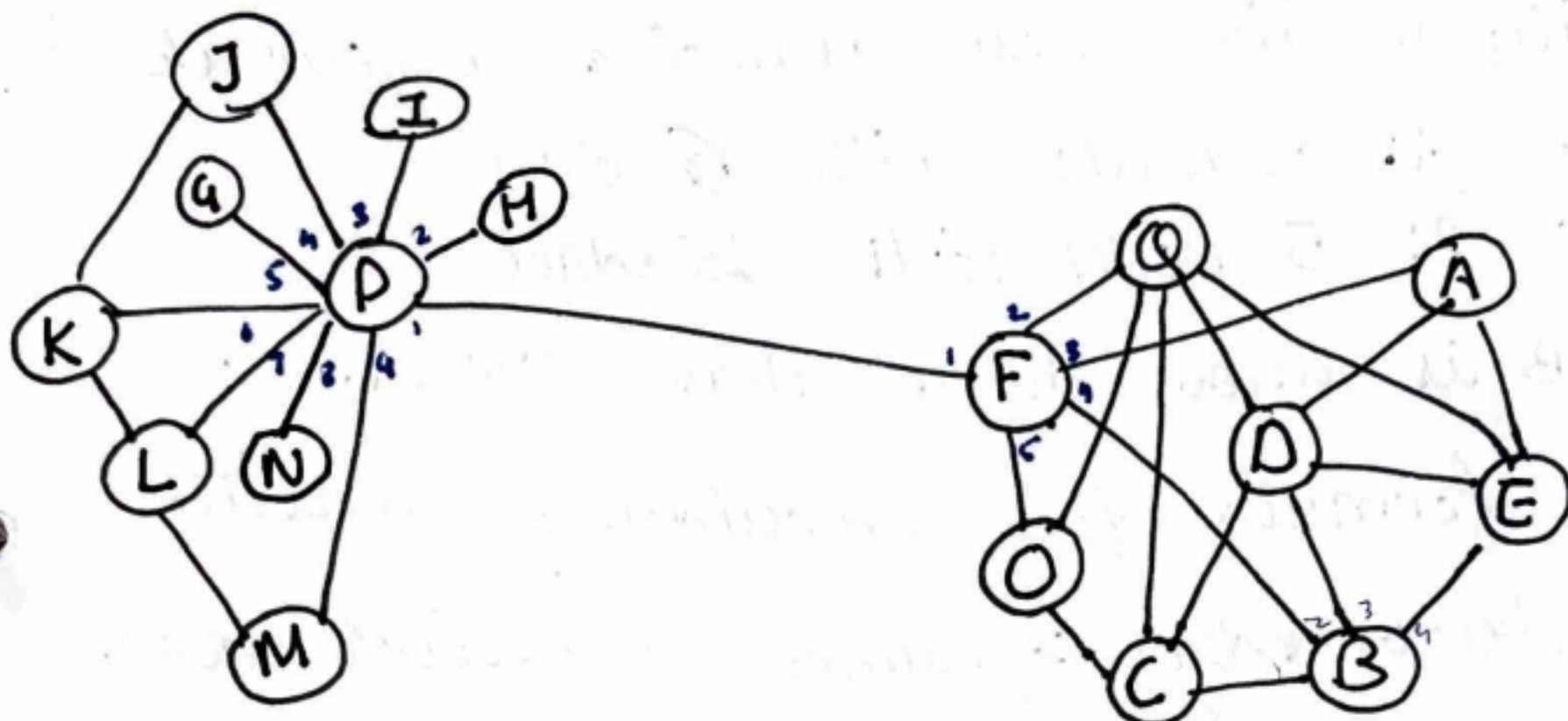
Degree distribution is a network measure that shows **how many nodes in a graph have a particular degree**. In simple words, it tells the frequency of nodes having 1 connection, 2 connections, 3 connections, and so on. It helps in understanding how the connections are spread in the network.

For example, if 4 nodes have degree 1, 2 nodes have degree 2 and 1 node has degree 3, then this frequency table is called the degree distribution of the graph.

 *Social Network Structure, Measures & Visualization*  Social Network Structure, Measu...

 *Adobe Scan – Page 1*  Adobe Scan 18 Mar 2025

Describing networks [density: $\frac{\text{no. of edges in graph}}{\text{no. of possible edges}}$]
 ↳ degree distribution.
 ↳ used to describe individual nodes.
 ↳



Node	Degree
1 A	3
2 B	4
3 C	4
4 D	5
5 E	4
6 F	5
7 G	1 (1)
8 H	1 (2)
9 I	1 (3)
10 J	2
11 K	3
12 L	3
13 M	2
14 N	1 (4)
15 O	3
16 P	9
17 Q	5

Degree distribution

Degree (1-9)	No. of Nodes
1	4
2	2
3	4
4	3
5	3
6	0
7	0
8	0
9	1

(17) Total nodes

iv) Write Business use case of Social Media.

One important business use case of social media is **marketing and promotion of products**. Companies use social media platforms such as YouTube, Facebook and Instagram to advertise products and interact with customers.

For example, Blendtec used the “**Will It Blend?**” campaign on YouTube to promote its blenders. This campaign became highly popular and significantly increased the company’s sales and brand visibility.

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v) Name the concepts to understand social media and business alignment.

The concepts used to understand how social media aligns with business objectives are:

- **Broadcast / Sending information** – sharing updates and promotions
- **Request / Feedback input** – collecting customer responses
- **Conversation interaction** – engaging in two-way communication

These concepts help businesses communicate effectively with their audience and align social media activities with business goals.

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vi) Write 3 tools used in Location and Search Analytics.

Three tools commonly used in location and search analytics are **Google Analytics, Google Search Console and Google Trends**.

Google Analytics helps track website visitors and their location, Search Console analyzes search engine performance, and Google Trends helps study keyword search trends over time.

These tools are useful for analyzing user behavior and search visibility.

 Relevant uploaded analytics resources  Chapter 06

vii) Explain with an example how social media is used by the government.

Government uses social media for **public communication, awareness and crisis management**. It helps in directly sharing updates, alerts and public notices with citizens.

For example, the **Philadelphia Police Department** used YouTube, Facebook and Twitter to post surveillance videos and emergency information, which helped identify suspects and solve cases.

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viii) Give Sources of Location.

The main sources of location are:

- **GPS (Global Positioning System)** – gives the exact geographic coordinates of the user.
- **IP Address** – provides the approximate location or region from which the user is accessing the internet.
- **Mobile network / Cell tower data** – identifies the user’s location based on the nearest network tower.
- **Wi-Fi network location** – estimates location using nearby Wi-Fi access points.
- **Geotagged social media posts / check-ins** – user-shared location information on social platforms.

 Chapter 06 – Privacy / data collection concepts  Chapter 06

ix) Define Search Engine Analytics.

Search Engine Analytics is the process of **analyzing search queries, keyword trends, click data and ranking performance** obtained from search engines.

It helps organizations understand what users search for and improve the visibility of websites in search results.



Search analytics concepts from uploaded resources



Chapter 06

x) Location Analytics has Privacy Concerns, Justify.

Yes, location analytics has privacy concerns because it collects **users' location, movement patterns and behavioral data**.

This information may reveal personal habits, frequently visited places and identity, which can be misused if privacy settings are not properly managed.



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Chapter 06

Q2 Theory Answers (5 Marks)

ii) Write and elaborate the steps to formulate social media strategy.

Formulating a social media strategy means creating a structured plan for how an organization will use social media to achieve its business goals.

1. Define objectives

The first step is to clearly define what the organization wants to achieve through social media. Objectives may include **brand awareness, customer engagement, lead generation, sales promotion, customer support, or community building**. Clear goals help in measuring success later.

2. Identify target audience

The organization must identify the **specific audience** it wants to reach, such as students, customers, professionals, or local communities. Understanding their **age, interests, location, behavior, and preferred platform** helps in designing relevant content.

3. Select suitable social media platforms

Different platforms serve different purposes.

For example:

- **Instagram / Facebook** → marketing and promotions
- **Twitter (X)** → updates and customer interaction
- **YouTube** → video campaigns
- **LinkedIn** → professional communication

The platform should be chosen based on business goals and target audience.

4. Create content strategy

A proper content plan should be developed which includes:

- posts
- videos
- stories
- campaigns
- feedback polls
- customer interaction posts

The content should align with business objectives and audience interest.

5. Engagement and communication plan

The business should decide how it will interact with users through:

- replies
- comments
- customer support
- live sessions
- feedback collection

This improves brand trust and engagement.

6. Measure and monitor performance

Success should be measured using:

- likes
- shares
- reach
- comments
- click-through rate
- conversions

Regular monitoring helps improve future strategies.

iii) Write risk management framework.

A risk management framework is a structured approach used to identify, assess, control and monitor risks associated with social media use.

1. Risk identification

The first step is identifying possible risks such as:

- privacy breach
- fake accounts
- misinformation
- negative publicity
- data theft
- reputation damage

This helps organizations understand what threats may occur.

2. Risk assessment

After identifying risks, the organization evaluates:

- how severe the risk is
- how likely it is to happen
- what impact it can create

This helps prioritize high-risk issues.

3. Risk mitigation / control

Control measures are applied to reduce risks, such as:

- strong privacy settings
- content moderation
- user access control
- security policies
- password protection
- two-factor authentication

 *Chapter 06 – Privacy and settings*  Chapter 06

4. Continuous monitoring

Risks on social media change quickly, so continuous monitoring is necessary. Businesses track:

- suspicious posts
- negative comments
- fake profiles
- unusual activities

This helps detect problems early.

5. Response and recovery plan

If a risk occurs, there should be a response plan such as:

- public clarification
- post removal
- issue resolution
- privacy recovery steps

This minimizes damage to brand image.

v) Explain Search Engine Analytics and the types of Search Engines.

Search Engine Analytics is the process of analyzing how users search information through search engines and how a website performs in search results.

It helps understand:

- which keywords users search
- which pages receive traffic
- ranking position
- click-through rates
- visitor behavior

This helps improve website visibility and SEO performance.

Importance

Search engine analytics helps businesses:

- increase online visibility
 - improve search ranking
 - understand user intent
 - optimize content
-

Types of Search Engines

1. Crawler-based search engines

These use automated bots (crawlers/spiders) to index websites.

Example: **Google**

2. Human-powered directories

These are manually categorized by humans into directories.

Example: **Yahoo Directory (older model)**

3. Meta search engines

These collect results from multiple search engines and display combined results.

Example: **Dogpile**

vi) Explain Business use of Social Media and how to measure success, interaction and monitor it.

Social media is widely used by businesses for communication, promotion and customer relationship management.

Business use of social media

Businesses use social media for:

- product marketing
- customer engagement
- customer support
- promotions and campaigns
- brand awareness

For example, Blendtec used the “**Will It Blend?**” campaign which greatly increased sales and brand recognition.

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Measuring success

Success is measured using performance indicators such as:

- number of views
- likes
- comments
- shares
- followers / subscribers
- conversions

These metrics indicate campaign effectiveness.

Interaction

Interaction refers to direct communication with customers through:

- replies
- comments
- chats
- customer service posts

Example: Zappos used Twitter for customer support.

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Monitoring

Businesses continuously monitor:

- customer feedback
- sentiment
- campaign reach
- complaints
- engagement trends

Monitoring helps improve future strategies and maintain reputation.

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